

CBSE Practice Worksheet

Class 6 | Mathematics | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$
(A) 21
(B) 23
(C) 25
(D) 27
2. Solve: $|2x - 5| = 9$
(A) $x = 7$ or -2
(B) $x = 7$ or 2
(C) $x = -7$ or 2
(D) $x = -7$ or -2
3. If $\log_{\blacksquare} 8 = x$, find x
(A) 2
(B) 3
(C) 4
(D) 8
4. Simplify: $(a^2 - b^2)/(a + b)$
(A) $a - b$
(B) $a + b$
(C) $a^2 - b$
(D) $a - b^2$
5. Solve: $\sqrt{x + 7} = x - 5$
(A) $x = 9$
(B) $x = 2$
(C) $x = 9$ or 2
(D) No solution
6. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$
(A) 3:6
(B) 5:8
(C) 15:24
(D) 1:2
7. If $2\blacksquare = 3\blacksquare = 6\blacksquare$, which is true?
(A) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$
(B) $z = x + y$
(C) $z = xy$
(D) $z = x - y$
8. Solve: $\frac{3(x-2)}{4} = \frac{(2x+1)}{3}$
(A) $x = 14$
(B) $x = 22$
(C) $x = 26$

(D) $x = 30$

9. Solve: $2^{x+1} = 32$

(A) $x = 3$

(B) $x = 4$

(C) $x = 5$

(D) $x = 6$

10. Factorize: $x^2 - 5x - 14$

(A) $(x-7)(x+2)$

(B) $(x+7)(x-2)$

(C) $(x-7)(x-2)$

(D) $(x+7)(x+2)$

Section B: Fill in the Blanks (1 mark each)

11. $\text{HCF} \times \text{LCM}$ of 12 and 18 = ____

12. If $x^2 - 6x + 8 = 0$, then $x =$ ____ or ____

13. If $5^x = 125$, then $x =$ ____

14. $\sqrt[3]{-64} =$ ____

15. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 -$ ____

16. If $a + b + c = 0$, then $a^3 + b^3 + c^3 =$ ____

17. If $\log x = 2$, then $x =$ ____

18. $(x + 1/x)^2 - (x - 1/x)^2 =$ ____

Section C: Short Answer Questions (2 marks each)

19. Factorize: $8x^3 - 27$

20. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

21. If $x = 2 + \sqrt{3}$, find $x + 1/x$

22. Solve: $|x - 3| + |x + 2| = 7$

23. Simplify: $(x^2 - 16)/(x^2 + 4)$

24. Find the value of x : $3^{2x-1} = 27^{x-2}$

25. Solve: $\log(x - 1) + \log(x + 1) = 3$

--- End of Worksheet ---

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